Emmanouil Giannakis

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EDUCATION

National Technical University of Athens (NTUA), Athens, Greece *Sept 2026 – Present*
MSc in Translational Engineering in Health and Medicine

University of Piraeus, Piraeus, Greece *Sept 2021 – Sept 2025*
BSc in Digital Systems, Majoring in Computational Infrastructures & Services
Thesis under the supervision of the Institute of Nuclear and Particle Physics at NCSR Demokritos:
"Neutrino's Event Reconstruction with Graph Neural Networks in ANNIE" **Grade: 10/10**
Significant Project:

- Designed and implemented a multi-agent system with dynamic task allocation, negotiation protocols, and performance evaluation in a collaboration-driven environment.

Lyceum of Ayia Paraskevi, Athens, Greece *2019 – 2021*
Honor Roll Student, High School Diploma Grade: **19.0 (GPA 4.0)**

Henry Hudson Elementary School, Vancouver, Canada *Sept 2013 – Jun 2015*
Studied at an international school and achieved native-level proficiency in English.

EXPERIENCE

Machine Learning Engineer (Freelance), Philon – Athens, Greece Aug 2026 – Oct 2026

- Built a per-frame video classification model from scratch, replacing an unevaluated legacy pipeline, hand-labelled a held-out test set so every model was benchmarked against the same ground truth.
- Trained via knowledge distillation from a prompted teacher model and replaced whole-frame mean pooling with region pooling, the largest single metric gain, the large model ($\sim$400M-parameter encoder) reached 0.895 precision.
- Delivered a compact model with a $\sim 5\times$ smaller encoder and a pipelined GPU inference path with no idle time, running $4.3\times$ faster ($7.37 \rightarrow 1.72$ ms/frame) at 0.926 accuracy and 0.840 precision.

Student Researcher, NCSR Demokritos – Athens, Greece Sept 2024 – Present

- Designing and fine-tuning GraphNet-based models for neutrino event reconstruction, leveraging training from similar experiments and analyzing results to guide model improvements (manuscript in preparation for submission to JINST).
- Co-developed and presented an open-source toolkit for converting ROOT files to SQLite, Parquet, and H5 formats, adopted in collaboration with the Fermilab team.

Head of Autonomous System, Prom Racing FSAE, NTUA – Athens, Greece Sept 2024 – Sept 2025
Technical:

- Familiarized with sensor calibration and fusion algorithms, including integration of data from IMU, INS, and camera-LiDAR systems, contributing to robust multi-sensor systems for autonomous navigation and environmental understanding.
- Designed and implemented custom datasets with augmentation for object detection and pose estimation tasks. Leveraged YOLO-Pose to unify both tasks and applied knowledge distillation, reducing inference time and enhancing precision with range detection.

Managerial:

- Secured a LiDAR sensor sponsorship, achieving a key milestone pursued for over three years.
- Oversaw two sub-team recruitments through a two-step selection process, integrating new members through technical and safety training.

Autonomous System Engineer, Prom Racing FSAE, NTUA – Athens, Greece Oct 2023 – Sept 2024

- Initiated R&D to evaluate the viability of computer vision projects and their investment potential.
- Customized a path planning algorithm inspired by RRT (Rapidly Exploring Random Trees) to fit specific requirements, integrating it into the ROS2 framework for enhanced autonomous navigation and decision-making.
- Presented the team's perception pipeline at a ZEISS competition, achieving 4th place out of 42 teams (top 10%).

SKILLS

Technical Skills: Computer Vision, Digital Image Processing, Machine Learning, Deep Learning, Knowledge Distillation, Model Compression, Sensor Fusion, Filtering and Calibration Algorithms, Environment Perception, Multi-Agent Systems, Probability Theory, Stochastic Processes, Model Evaluation, Data Engineering and Visualization, and Testing.
Interpersonal Skills: Teamwork, Communication, Project Management, Presentation, Public Speaking, Problem Solving.
Languages: English (Proficient), Greek (Native).
Programming and Tools: Linux, Git, Bash, ROS2, Python, C, C++, Java, HTML, MATLAB, SQL.

CERTIFICATIONS

Volunteering Assistant, HIAS AI Summer School, NCSR "DEMOKRITOS" July 2024
Fundamentals of Deep Learning, NVIDIA May 2024
Unmanned Aerial Systems Common Module, Hellenic Air Force Apr 2024
Universal AI Course, NCSR "DEMOKRITOS" July 2023